

DESIGN CHALLENGE GENERAL SPECIFICATION AND RULES.

Regional Competitions – UNITED ARAB EMIRATES



Undergraduate Engineers Regional Competition

**This Specification must be read in
Conjunction with the Appropriate Project
Specification**

1. Introduction and purpose of the Design Challenge

This specification and rules of engagement are to be used in conjunction with the specific project specification adopted at the time.

The purpose of the Design Challenge competition is to simulate a real-world project, allowing students to develop cross-functional skills. Working in teams, students will design, build and test a self-contained device from a precise Project Specification. The competition is open to teams of **two to five engineering** degree students studying at the relevant academic level.

Each team can use just **one device** at the competition but all students who work on the challenge will benefit from the learning experience of applying their engineering knowledge.

The Challenge consists of four elements which are.

- 1) 1a - Work in teams to design, build and test a self-contained device from a precise specification. From this specification they have to produce a design solution, make it, test it and compete to win a place in the IMechE Regional Competition.
- 1b - Compete with other teams in the regional competition to achieve the fastest or most accurate device.
- 2) Produce a "poster" to demonstrate pictorial and graphical skills and the team's ability to sell their design solution.
- 3) Deliver a presentation using **a laptop** to demonstrate verbal and presentational skills. The digital CAD model and Bill of Materials will be used to assess the technical competence of the presented device.
- 4) Carry out a "peer review" of the devices produced by all of the other teams and rate them so that a winner can be determined.

The above covers the general requirements of a professional engineer.

The four sections of the regional DC competition are as follows.

- 1) Main challenge competition.
- 2) Presentation competition.
- 3) Poster competition.
- 4) Peer review.

Rules for the poster competition.

See the poster judging criteria in appendix - B

1. The poster should be A1 size in portrait format. It should clearly display the logos of the team's university and of the IMechE.
2. The poster is a demonstration of the team's ability to communicate their design solution, the journey to achieving the design, display of cost and performance graphically.

- 2.1 Sketch, 3D visualisation or 2D technical drawings
- 2.2 Text to introduce the competition and describe the device
- 2.3 Engineering calculations with expected – calculated performance
- 2.4 Total BoM cost, mass

3. The poster will be assessed and judged by the appointed judges. In accordance with the marking scheme in appendix "B"
4. The poster is a demonstration of the team's ability to sell their design solution.

Rules for the presentation competition.

See the presentation judging criteria in appendix – C

The purpose of the presentation is to explain to an audience of any background and understanding, the purpose of the product, its design from the initial concept selection through the evaluations, the expected and calculated performance cost.

The run order of the presentations will be determined on the day of the competition.

1. Presentations should be submitted (on a memory stick) on arrival at the regional competition.
2. The maximum length of the presentation is five minutes plus typically two minutes for questions. It can be delivered by any number of team members, from one person to all members of the team. Computer and projector facilities with common software will be available.
3. The presentation should include, but is not limited to;
 - 3.1. Introduce the design challenge and describe briefly the final competing device
 - 3.2. the engineering science that underpins the device and steps the team followed to arrive at the design,
 - 3.3. the cost of the final design and if/how costs influenced the final design and CAD mass
4. The team will be required to answer questions on their design.
5. The presentation will be assessed by the judges according with the marking criteria in the "Judges procedure appendix C, and will be judged by the appointed judges.

Rules for the peer review competition.

See the peer review voting slips in appendix – D

The peer review purpose is the opportunity for students to express their view on the design and execution proposed by other teams.

1. Each team should examine the device design from each of the other teams without handling them.
2. Whilst the peer review is being carried out there must be at least one member of each team present to answer questions etc.
3. During the examination teams should be looking for the following;
 - 3.1 design principles used,
 - 3.2 the simplicity of the design,
 - 3.3 the robustness of the design,
 - 3.4 the manufacturing excellence, and
 - 3.5 the general appearance.
4. The competition judges should cast a cursory eye over the procedure during the peer review.

2. Enforcement of the Rules.

1. On matters relating to test equipment and procedure, the authority will be the chair of the Institution of Mechanical Engineering Design Challenge organising committee or his/her delegated representative(s).
2. The panel of judges consists of IMechE and university/industry representatives.
3. The decisions of the panel of judges will be final.
4. In addition to the rules for the regional competition outlined above, universities are responsible for internally ensuring that the spirit of the competition is adhered to during the design and make stages.
5. Appeals: If a team wishes to lodge a complaint to query a procedure or rule infringement, they must do so through the chair of the institution of Mechanical Engineering Design Challenge organising committee or his/her delegated representative(s). Any complaint will be investigated immediately with at least two judges and a response will be issued within a reasonable time. This decision will be final and not subject to further appeal.

3. Judges procedure and scoring.

Judges will be appointed by IMechE and these would normally be the representatives from the regional Design Challenge Committee and from the IMechE.

Note – The judging panel are allowed to vary the rules slightly if it is deemed necessary to maintain the smooth running of the competition.

The Judge's decision will be final.

4. Prizes and Certificates.

- a) Cash prizes AED 1500 will be awarded to the winning team in the Final.
- b) There will be separate, independent, prizes for the poster, and peer review elements of the challenge.
- c) The winning team's institution shall be awarded a rolling trophy
- d) Appropriate certificates of recognition will be awarded to all students and the winning institute.

Appendix A –Run Order

(Scores of Heats to be displayed prior to the start of the main competition).

	Heat 1	Heat 2	Heat 3	Heat 4	Heat 5	Heat 6	Etc. etc.
University							
University							
University							

Appendix B - Poster Judging Criteria

		<u>Weight (%)</u>
Visual Impact	Compliance with rules – size (A1) and orientation (portrait)	10
	Obvious information on the university represented (logos) and the team members' names	10
	Good use of colour, readable. Units and decimals appropriate. No grammar or spelling errors.	15
Technical Content	Clear but brief textual description of the competing device	20
	Clear diagram(s) – sketch, rendering or CAD model – of the device	15
	Evidence of the engineering science underpinning the device	15
	Different BoMs, summary costs of the different elements, mass	15
	<u>Total</u>	<u>100</u>

Appendix C - Presentation Marking Scheme

		<u>Weight (%)</u>
Presentation Style	Overall quality of presentation	15
	Quality and integrity of visual aids and content	15
Technical Content	Introduction of the competition, summary, team details	15
	Steps followed to reach the final design, BoM analysis	15
	Engineering science that underpins the final design	20
Review at competition	Answer to judges questions	20
	Max Points	100